Internship Report

Of

AUTOMATION TESTING OF SAUCEDEMO WEBSITE USING SELENIUM WITH PYTHON

Submitted
To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE

By

Roll Number of Student: 2410031285

Name of Student: ANKIT KUMAR DUBEY

Batch: 2024-28

CANDIDATE'S DECLARATION

I, Ankit Kumar Dubey, do hereby declare that the internship report titled "Automation Testing of SauceDemo Website Using Selenium with Python" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science & Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _______________

Student Name: Ankit Kumar Dubey

Roll No.: 2410031285

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature: ______________

Student Name: Ankit Kumar Dubey

Roll No.: 2410031285

INTERNSHIP COMPLETION CERTIFICATE

Sopra Steria India Limited
(Formerly known as Steria (India) Limited)
Seaview Special Economic Zone, Building 4
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Gautam Buddh Nagar, Noida (UP) 201304, India
Tel. +91-120-302 1330
www.soprasteria.in

Date : 21st August 2026

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Ankit Kumar Dubey successfully completed his project from 15/06/26 to 21/07/26 for the partial fulfillment of B.Tech. The project titled **"AUTOMATION TESTING OF SAUCEDEMO WEBSITE USING SELENIUM WITH PYTHON"** was done at Sopra Steria India, under the guidance of Anmol Dogra.

We congratulate you on the successful completion of the project and wish you success in all future endeavors!

Mentor

Name: Anmol Dogra

Designation: Technical Lead-Testing

Anmol

Approving Authority

Name: Chanchal Chauhan

Designation: Head - Talent Development

Chanchal Chauhan

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4. PROJECT DESCRIPTION

4.1 Introduction

Software testing is an essential phase of the Software Development Life Cycle (SDLC). Its primary objective is to verify that software behaves as expected and meets user requirements. Traditionally, software testing was performed manually; however, with increasing application complexity and frequent software releases, automation testing has become a necessity. Automation testing improves software quality by executing repetitive test cases with greater speed, accuracy, and consistency compared to manual testing, while also reducing human intervention and minimizing the possibility of execution errors.

The objective of this internship project was to automate the testing of the SauceDemo e-commerce website using Selenium WebDriver with Python. The application provides various functionalities such as user authentication, product browsing, sorting, shopping cart management, and checkout. Automating these functionalities helps validate the application's behaviour under different scenarios. The automation framework developed during this internship follows the Page Object Model (POM), where each webpage is represented as a separate module, improving code readability, simplifying maintenance, and allowing better scalability for future enhancements.

4.2 Organization Profile

Sopra Steria is one of Europe's leading technology consulting and digital transformation organizations. The company delivers end-to-end business solutions by combining consulting, software development, systems integration, cloud technologies, cybersecurity, data analytics, artificial intelligence, and quality engineering services.

Operating across multiple countries, Sopra Steria serves clients from industries such as banking, insurance, aerospace, defence, government, transportation, retail, manufacturing, and telecommunications. The organization focuses on helping businesses accelerate digital transformation while maintaining high standards of innovation, reliability, and customer satisfaction.

The company promotes continuous learning, technical excellence, teamwork, and innovation, and through its internship programs, students gain practical exposure to modern software engineering practices and industry-standard development methodologies.

During this internship, the author worked on an automation testing project using Selenium WebDriver with Python. The project provided an understanding of real-world software testing processes, automation framework design using the Page Object Model (POM), assertions, browser automation, and structured project development, significantly improving technical knowledge, analytical thinking, debugging ability, and understanding of software quality assurance practices.

4.3 Problem Statement

Manual testing of a web application such as SauceDemo becomes repetitive, time-consuming, and prone to human error when the same set of test cases — login validation, product verification, sorting, cart management, checkout, and logout — needs to be executed repeatedly across releases. There was a need for a reliable, repeatable, and efficient mechanism to validate these core business functionalities without manual intervention every time the application changed. This project addresses that need by building an automation framework that simulates real user interactions and validates the expected outcomes automatically using Selenium WebDriver with Python.

4.4 Project Objectives

The primary objective of this project was to automate the functional testing of the SauceDemo e-commerce web application using Selenium WebDriver with Python, eliminating repetitive manual activities and ensuring accurate execution of predefined test cases. The project focuses on validating the major functionalities of the application, including:

- User authentication (valid and invalid login scenarios)
- Product verification and inventory inspection
- Product sorting/filtering
- Shopping cart operations
- Checkout process and order completion

- Logout functionality

Each automated test executes user actions in the same manner as a real user and verifies the expected outcome using assertions.

4.5 Scope of the Project

The scope of this project is limited to functional automation testing of the SauceDemo web application. The automation scripts verify whether the application behaves correctly under different user scenarios. The implemented automation covers the following core features:

- Successful Login and Invalid Login Validation
- Product Information and Inventory Verification
- Product Sorting using Filter Options (A-Z, Z-A, Price Low-High, Price High-Low)
- Shopping Cart Operations (adding items and validating contents)
- Checkout Process (entering customer information and reviewing overview)
- Order Completion and confirmation validation
- Application State Management (Reset App State)
- Logout Functionality and Verification

Future Extension Scope: The project can be extended in the future to support cross-browser automation, data-driven testing (via Excel/CSV), parallel execution, automated reporting tools, and CI/CD pipeline integration.

4.6 Technologies and Tools Used

Hardware Requirements:

Component	**Specification**
Processor	Intel Core i5 or higher
RAM	Minimum 8 GB (16 GB recommended)
Storage	Minimum 256 GB SSD
Operating System	Windows 11
Internet Connection	Required

Software Requirements:

Software	Purpose
Python 3.x	Programming language
Selenium WebDriver	Browser automation tool
Visual Studio Code	Integrated Development Environment (IDE)
Google Chrome	Target web browser
ChromeDriver	Browser driver bridge

4.7 System Architecture

The automation framework follows a strict layered architecture based on the Page Object Model (POM) pattern. Test scripts interact only with page classes rather than directly accessing web elements, establishing a clean separation of concerns. The execution flow proceeds from the test execution script, through Selenium WebDriver, to the Google Chrome browser, and finally to the SauceDemo website, with the Page Object Model layer (Login, Inventory, Cart, Checkout, and Menu pages) sitting between the tests and the browser.

Project Folder Structure: the project is organized modularly to maximize maintainability, scalability, and reusability, as summarised in Table 4.1.

Directory / Folder	Files Included	Purpose
pages/	login_page.py, inventory_page.py, cart_page.py, checkout_page.py, menu_page.py	Contains Page Object Model (POM) classes defining page elements and actions
tests/	test_complete_flow.py, test_invalid_login.py	Contains actual test script execution files and assertions
utilities/	browser.py	Handles browser initialization, driver configuration, and teardown setup
test_data/	credentials.py	Safely stores user login inputs and test datasets
Root Directory	requirements.txt, README.md	Lists external Python project dependencies and basic usage instructions

Table 4.1: Project folder structure

4.8 Methodology

The developed automation framework simulates the complete workflow of a customer interacting with an e-commerce website. Instead of manually performing each activity, Selenium WebDriver automatically controls the browser and performs all operations sequentially. The project execution flow is as follows: (1) Launch & Authentication — the framework launches Google Chrome, opens the SauceDemo website, and enters valid user credentials; (2) Inventory Inspection — after successful login, the script verifies the product inventory page and retrieves product names, prices, and descriptions; (3) Filter Sorting — the framework validates the product filter functionality by executing all available sorting options; (4) Cart & Checkout Transaction — a product is added to the shopping cart, details are validated, and the script proceeds to the checkout page to insert simulated customer data; (5) State Reset & Teardown — once the order confirmation is asserted, the framework resets the application state, logs the user out, and safely closes the browser.

4.8.1 Module Description

Browser Utility Module: launches the Google Chrome browser, maximizes the browser window, applies browser configurations, and closes the browser after execution, providing a centralized location for browser initialization and reducing duplicate code.

Login Module: automates user authentication by entering valid username and password credentials into the login page. After clicking the Login button, assertions verify whether the user is redirected to the Inventory page successfully.

Key tasks: open website, enter username/password, click login, validate success.

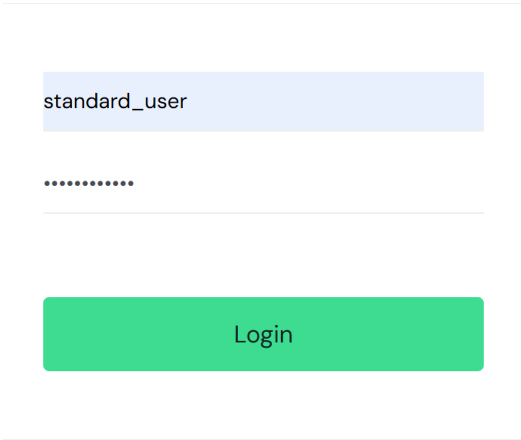

Figure 4.1: Successful Login Page

Invalid Login Module: validates the application's behaviour when incorrect login credentials are entered. The framework verifies that the appropriate error message is displayed and confirms that unauthorized access is prevented.

Key tasks: enter invalid credentials, verify error UI message elements.

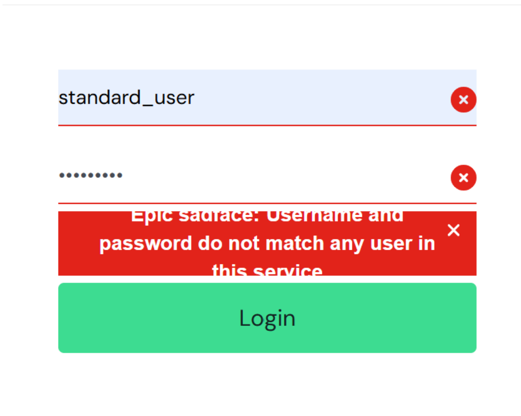

Figure 4.2: Invalid Login Validation

Inventory Module: after successful login, retrieves and validates available product information displayed on the application, automatically extracting product names, descriptions, and prices.

Key tasks: console printing and data verification of products.

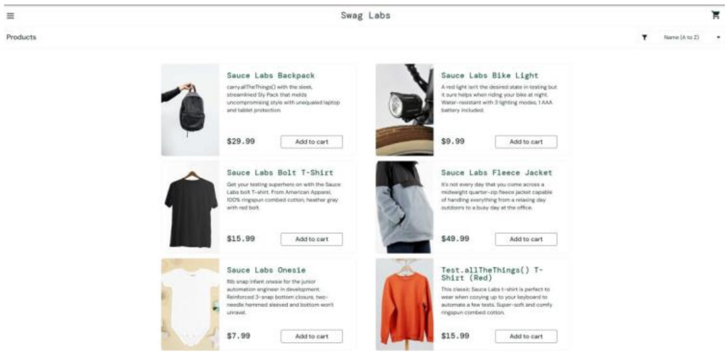

Figure 4.3: Inventory Page with Product Details

Product Filter Module: validates all sorting options available on the inventory page, executing Name (A to Z), Name (Z to A), Price (Low to High), and Price (High to Low) filters sequentially.

Key tasks: change dropdown options, verify structural rearrangement of items.

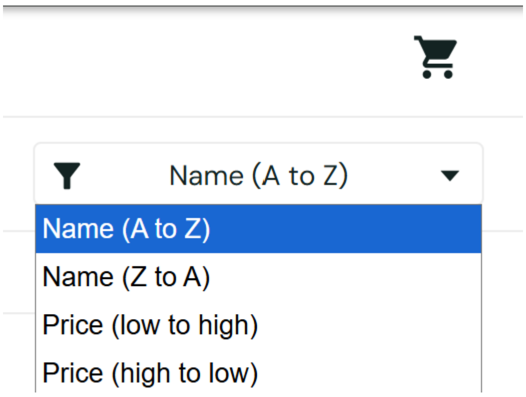

Figure 4.4: Product Filter Validation

Cart Module: validates the shopping cart functionality after product selection. Once a product is added, the script navigates to the cart and verifies product name, description, price, and quantity using assertions to prevent data inconsistency.

Key tasks: match selected item properties with cart item properties.

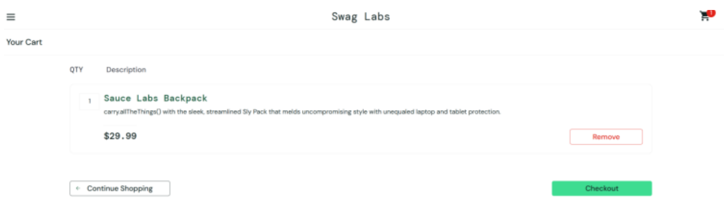

Figure 4.5: Shopping Cart Page

Checkout Module: automates the purchasing process by entering customer information and completing the order. It inputs first name, last name, and postal code, then passes through the checkout overview page to finalize the purchase.

Key tasks: enter customer metadata, execute checkout, assert final success message.

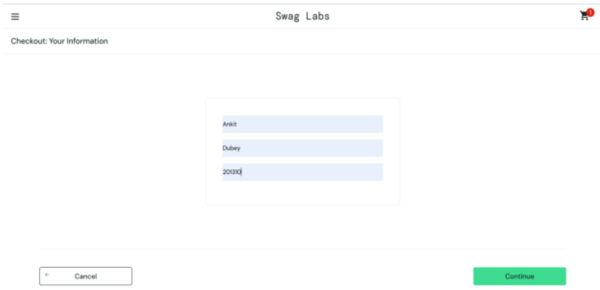

Figure 4.6: Checkout Information Page

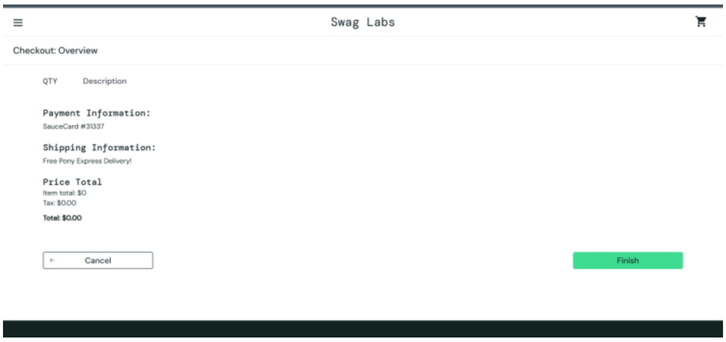

Figure 4.7: Checkout Overview Page

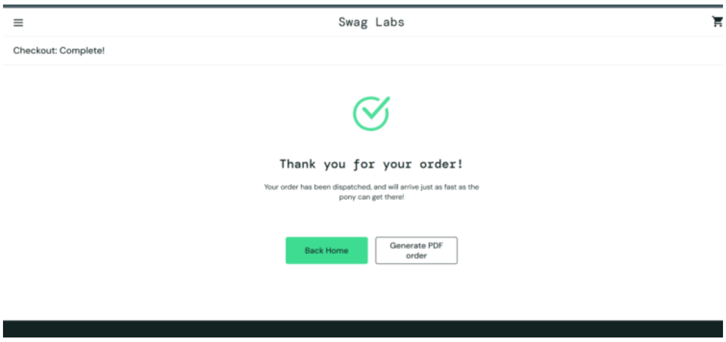

Figure 4.8: Order Confirmation Page

Menu Module: automates operations available from the side navigation menu after login. It manages opening the side menu drawer, executing the “Reset App State” option to clear cart memory, and logging out to return to the root screen safely.

Key tasks: application cleanup, user logout action validation.

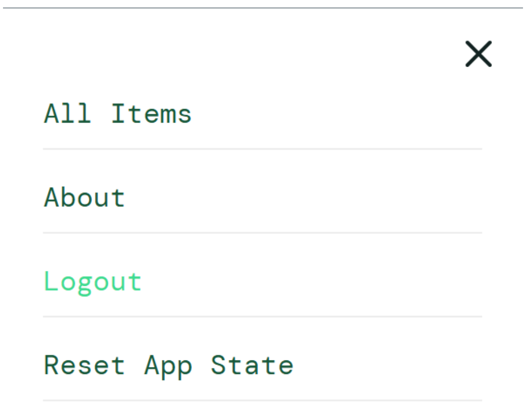

Figure 4.9: Application Menu Options

4.8.2 Assertions Used

Assertions form the foundational validation framework of the script. Table 4.2 lists the checks executed to confirm expected versus actual outcomes.

Validation Target	**Assert Condition / Expected Result**
Login Validation	URL/page title contains “inventory.html” after valid form submission
Shopping Cart Validation	Cart badge counter increments and equals the number of chosen products
Cart Verification	Product name, price, description, and quantity exactly match selected values
Checkout Validation	Submission moves the flow forward into the Checkout Overview page
Order Completion	“THANK YOU FOR YOUR ORDER” header text is present and displayed
Logout Validation	Application correctly returns the user context back to the root login screen

Table 4.2: Assertions used in the project

4.9 Expected Outcomes

The completed automation framework successfully automated the end-to-end user workflow on the SauceDemo site. The framework ran reliably without user intervention, successfully completing all operations: authenticating users, catching invalid logins, matching prices, validating filtering logic, purchasing targeted cart products, confirming order completion headers, and tearing down states cleanly via automated logouts. Every assertion in the test scripts passed successfully, demonstrating that automation with Selenium and Python cuts operational time down to a fraction of manual overhead.

Applications of the Project:

- Regression Testing — automatically re-running the suite after system updates to ensure core functions are unbroken.
- Smoke Testing — rapidly executing basic operational checks (such as verifying login works) before broader manual tests.
- CI/CD Pipeline Integration — can be bound directly into build tools to run unattended on every deployment.
- Efficiency Gains — drastically cuts execution time and avoids the human error common in manual e-commerce path validation.

Challenges Faced:

- Browser Driver Configuration — initial ChromeDriver/Selenium compatibility issues were resolved by matching the driver version exactly to the installed Google Chrome version.
- Element Identification — dynamic attributes on standard elements required advanced selector inspection; resolved by prioritizing fixed IDs, explicit structural XPaths, and robust CSS selectors.
- Synchronization Issues — asynchronously loaded elements during fast page transitions caused execution exceptions; resolved by implementing precise Selenium dynamic wait strategies.
- Project Organization — rising code complexity as pages grew was resolved by strictly following the Page Object Model (POM).

Conclusion: the automation testing of the SauceDemo website using Selenium WebDriver with Python was successfully completed during the internship at Sopra Steria Group. The framework demonstrated how automated scripts reduce repetitive overhead while improving test suite accuracy across multiple operational workflows. Following the modular Page Object Model (POM) pattern keeps the repository readable and ready for future updates, and the real-world quality assurance skills gained through this industry project provide a strong foundation for professional development.

Future Enhancements: to move this automation setup toward enterprise grade, future iterations can introduce:

- Advanced reporting utilities such as PyTest HTML, Allure, or Extent Reports with embedded error screenshots.
- Data-driven testing using external Excel or CSV file integration to test many user account inputs.
- Multi-browser parallel execution (Chrome, Firefox, Safari, Edge) via Selenium Grid.
- Full CI/CD pipeline orchestration using GitHub Actions or Jenkins.

4.10 Certificates of Completion and Other Communication Mail Proof

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C2 - Restricted use

sopra steria

Subject: Offer of Remote Internship

Dear Ankit Dubey,

Congratulations!
We are pleased to offer you a Remote Internship with Sopra Steria India Ltd.

This opportunity is subject to your acceptance of the terms below and submission of the documents listed in Annexure I.

Remote Internship Terms & Conditions:

a. The duration of the Remote Internship shall not be less than 4 weeks and shall not be more than 12 weeks.
b. You may be assigned an “Idea Project”, which may not necessarily align with your academic specialization or area of interest.
c. You will not have regular access to company premises or infrastructure. Visits, if there are any, will be limited and require prior approval.
d. In case, the mentor requires an in-person meeting, the Intern will be allowed only under the mentor’s physical supervision and responsibility. The Mentor must obtain prior approval from the Talent Development SPOC for the same.
e. Any kind of storage device such as Laptops, USBs, or other external computing device must be declared at the Sopra Steria India’s security check point and are only permitted up to the reception area.
f. Sopra Steria India does not charge anything and does not pay any stipend for Remote Internship. Also, no travel or accommodation reimbursement is provided.

Remote Internship Project Completion Requirements:

a. Submit the soft copy of your project report (and executable file, if applicable).
b. You may be asked to present the project.
c. Upon the approval of the mentor, your Remote Internship Certificate will be issued within 15 working days. Talent Development reserves the right to review the project and decide the completion.

Annexure I – Required Documents (Scanned Copies):

- Aadhar Card (as ID proof)
- Photograph: One (1) scanned photograph

Chandan Chauhan

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Sopra Steria India Limited
(Formerly known as Steria (India) Limited)
Seaview Special Economic Zone, Building 4
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Date : 21st August 2026

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We congratulate you on the successful completion of the project and wish you success in all future endeavors!

Mentor

Name: Anmol Dogra

Designation: Technical Lead-Testing

Anmol

Approving Authority

Name: Chanchal Chauhan

Designation: Head - Talent Development

Chanchal Chauhan

Remote Internship - Offer Letter External Inbox ×

TYAGI Aashish <aashish.tyagi@soprasteria.com>
to me

Fri, Jul 3, 10:13 AM

Hello Ankit,

As discussed during your call with your mentor, your remote internship with us is subject to your formal acceptance of the Offer Letter.

Please find attached the offer letter detailing the terms and conditions of your remote internship. Kindly go through the document carefully and share your acceptance by replying to this email.

The details of your remote internship project will remain the same as finalized during your discussion with your mentor. Additionally, you would have received the format of project report from your mentor. In case you have not received it yet, please connect with your mentor directly and request them to share the sample format.

Post completion of your project, here is the process for issuance of the Remote Internship Certificate:
- The intern must first email the soft copy of the final report to the assigned mentor for content approval.
- Once the mentor approves the report, they will share it with us through the MS Forms submission process along with confirmation for certificate issuance.
- As mentioned in the offer letter, the certificate will be issued within 15 - 20 working days after final submission and approval.

(If your project involves coding, kindly prepare a separate document containing the complete project code and share it in a zipped folder at the time of submission.)

We wish you a successful learning journey with us.

Thank you!

5. BIBLIOGRAPHY/REFERENCES

References

- Selenium WebDriver Official Documentation, available at: selenium.dev/documentation.
- Python Official Language Documentation and Standard Libraries.
- SauceDemo E-Commerce Practice Environment, available at: saucedemo.com.
- Visual Studio Code Environment Guide Documentation.
- YouTube and W3Schools Python Automated Testing Guides.

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- Official ChromeDriver Configuration and Systems Communication Documentation.
- Open-source automation communities and the Stack Overflow developer archive.